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8. Worked example

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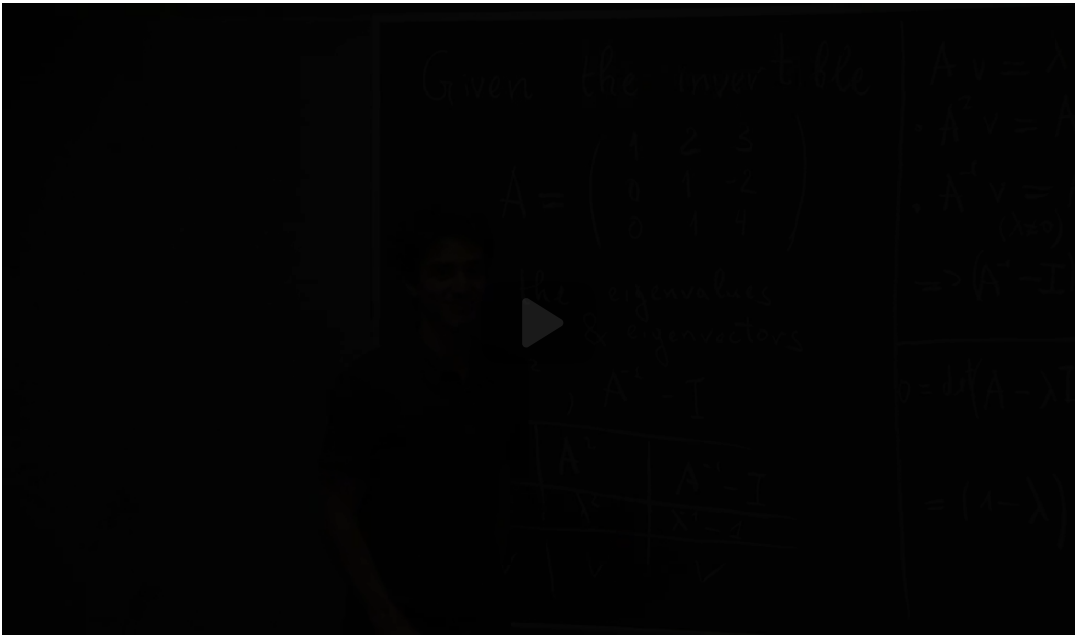
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Eigenvalues and Eigenvectors



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So A^2 minus the identity.
So the first row will be eigenvalues.
So it's going to be λ 's an eigenvalue for A ,
then we saw that λ^2 will be the eigenvalue for A^2 , and λ^{-1} will be the eigenvalue for A^{-1} minus the identity.
And the eigenvectors will be the same.
OK, we're done.

Eigenvalues and Eigenvectors Concept Check

1/1 point (graded)

Suppose that $\mathbf{v}$ is an eigenvector of a matrix $\mathbf{A}$ with associated positive real eigenvalue λ . Is $\mathbf{v}$ an eigenvector of $\mathbf{A}^3$? If so, what is the eigenvalue of $\mathbf{v}$?

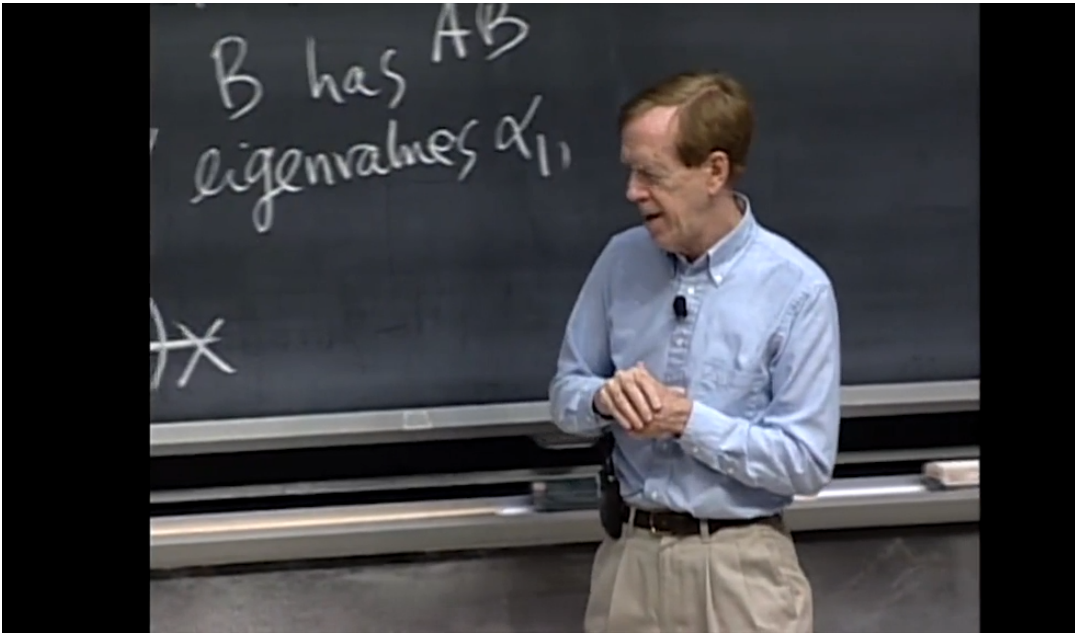
- ☐ No, $\mathbf{v}$ is not an eigenvector of $\mathbf{A}^3$
- ☐ Yes, $\mathbf{v}$ is an eigenvector of $\mathbf{A}^3$ with eigenvalue $\sqrt{\lambda}$.
- ☐ Yes, $\mathbf{v}$ is an eigenvector of $\mathbf{A}^3$ with eigenvalue λ .
- ☒ Yes, $\mathbf{v}$ is an eigenvector of $\mathbf{A}^3$ with eigenvalue λ^3 .



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Caution



of $A + B$
or $A \times B$ are not eigenvalues of $A + B$ plus eigenvalues of B .
Eigenvalues are not, like, linear.
And they don't multiply.
Because the eigenvectors are usually different,
and there's just no way to find out what $A + B$ does to a vector.
OK, so that's, like, a caution.
If B is a multiple of the identity, great.
But if B is some general matrix, then, for $A + B$,
you've got to solve the eigenvalue problem.

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